

Minimum Variance Delta Hedging

Project Summary

This project studies hedge-ratio misspecification in delta hedging, which can arise when implied volatility co-moves with the underlying asset price and creates residual volatility exposure. Standard Black-Scholes delta hedging assumes constant volatility and continuous rebalancing. In practice, implied volatility changes with the underlying asset price, and hedging is performed discretely. As a result, Black-Scholes delta hedging can leave residual hedging error.

The project implements a minimum-variance delta hedging framework inspired by Hull and White's research. It uses a synthetic SPY option panel built from historical SPY prices, VIX as an implied-volatility anchor, and a quadratic log-moneyness volatility surface.

The purpose is to estimate a correction to standard Black-Scholes delta by incorporating the predictable component of implied volatility changes that co-move with the underlying asset price.

The pooled panel regression produces stable coefficient estimates after the initial burn-in period. The minimum-variance delta reduces squared hedging error by approximately 51% on the raw rolling sample and 63% after applying the burn-in filter. These Gains are materially higher than those reported in Hull and White's research, suggesting that the synthetic data-generating process is favorable to the minimum-variance hedge. The centered-basis specification improves numerical stability without materially changing the fitted correction curve.

Delta-bucket analysis shows positive Gains across most delta regions, while the extreme high-delta bucket is unstable and excluded from the trimmed evaluation. The pattern of Gains across delta buckets differs from Hull-White's empirical results. The fitted correction curve is negative across the delta range and approximately U-shaped, with the largest correction around the central delta region.

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Black-Scholes delta hedging does not work perfectly in practice. The delta-hedged option portfolio may have residual error for at least two broad reasons: one is hedge-ratio misspecification and another is discrete rebalancing. These two residual risks are complementary.

As for the hedge error regarding inappropriate hedging ratio, it derives from volatility-price correlation, which brings unhedged volatility exposure. This hedge error is systematic and predictable, and can be absorbed through minimum-variance delta into the underlying position. However, the discrete hedging error is resulted from non-continuous hedging. Under the assumption of the continuous rebalancing in Black-Scholes, delta hedging removes the instantaneous underlying randomness perfectly, so the hedged portfolio is instantaneously riskless, whereas in discrete time, the cancellation is no longer exact, so residual random P&L appears, and residual risk accumulates. Discrete hedging error per-period is approximately $\frac{1}{2}\sigma^2 S^2 \Gamma dt (\phi^2 - 1)$. ϕ^2 is Chi-squared under normality. So, discrete hedging error is Gamma P&L randomness, and is irreducible for a given dt . It needs to analyze the optimal rehedge intervals, choosing optimal timing to re hedge based on realized gamma.

Here, this project will focus on the first hedging error, and analyze the minimum-variance delta to minimize variance of error. Since these two hedging errors are orthogonal, fixing the hedge ratio does not remove discrete hedging error, and it cannot neutralize the nonlinear exposure perfectly during the discrete rebalancing interval.

1. Theoretical Formula

According to Black-Scholes theory, Δ_{BS} hedge for stock price movements, with assumption that volatility stays constant. Therefore, the delta hedging eliminates all diffusion risks under the assumption of constant volatility. However, the implied volatility changes associated with the underlying asset. Delta hedging leaves residual variance, with the assumption that volatility is stochastic. The minimum variance delta Δ_{MV} is the hedge ratio that minimizes the variance of the hedging error, $dV - \Delta_{MV}dS$, and it improves upon the traditional Δ_{BS} by incorporating implied volatility movement varying with the underlying asset.

V is the option price observed from market, and based on the Black-Scholes formula, we can write $V = f_{BS}(S, \sigma_{imp})$

Construct a delta hedged portfolio: $\pi = V - \Delta S$

The change of portfolio across one time-step dt : $d\pi = dV - d\Delta S = dV - \Delta dS$, holding Δ fixed across dt

$$dV = \frac{\partial V}{\partial S} dS + \frac{\partial V}{\partial \sigma_{imp}} d\sigma_{imp} + O(dS^2)$$

$$dV = \Delta_{BS} dS + vega_{BS} d\sigma_{imp} + \epsilon$$

$$d\pi = (\Delta_{BS} - \Delta) dS + vega_{BS} d\sigma_{imp} + \epsilon$$

If volatility were constant, $d\sigma_{imp} = 0$; if with stochastic volatility, $d\sigma_{imp} \neq 0$, then residual variance remains via $vega_{BS}$.

Next, choose Δ to minimize $Var(d\pi)$, namely minimum variance delta $\Delta_{MV} = \arg \min_{\Delta} Var(d\pi)$. This optimality is local in time, namely the hedge ratio minimizes the variance of the hedging error over an infinitesimal time interval, rather than achieving global replication over the option's life.

$$\begin{aligned} Var(d\pi) &= Var((\Delta_{BS} - \Delta) dS + vega_{BS} d\sigma_{imp}) \\ &= (\Delta_{BS} - \Delta)^2 Var(dS) + vega_{BS}^2 Var(d\sigma_{imp}) + 2(\Delta_{BS} - \Delta) vega_{BS} Cov(dS, d\sigma_{imp}) \end{aligned}$$

$$\text{Set } \frac{\partial Var(d\pi)}{\partial \Delta} = 0, \Delta_{MV} = \Delta_{BS} + vega_{BS} \frac{Cov(dS, d\sigma_{imp})}{Var(dS)}$$

$$\text{As a Local optimal hedging, } \frac{Cov(dS, d\sigma_{imp})}{Var(dS)} = \frac{E[d\sigma_{imp}]}{dS}$$

$$\text{Therefore, } \Delta_{MV} = \Delta_{BS} + vega_{BS} \frac{E[d\sigma_{imp}]}{dS} \quad (1)$$

$\frac{E[\epsilon]}{dS}$ is the residual term. When S follows a diffusion process, with no jumps, $dS \rightarrow 0$, then $\frac{E[\epsilon]}{dS} \rightarrow 0$, namely higher-order terms vanish in the local asymptotic expansion.

$vega_{BS} \frac{dE[\sigma_{imp}]}{dS}$ is a correction term, reflecting volatility dynamics.

2. Empirical Analysis Formula

As for empirical analysis, we introduce an assumption of empirical structure for $\frac{dE[\sigma_{imp}]}{dS}$ (Hull&White, 2017)

$$\frac{dE[\sigma_{imp}]}{dS} = \frac{1}{S_t\sqrt{T}}(a + b\Delta_{BS} + c\Delta_{BS}^2)$$

$\frac{1}{S_t}$ guarantees the scale invariance under S, since any sensible hedging rule must be scaled invariant.

$\frac{1}{\sqrt{T}}$ enforces time consistency.

Empirical observations (Hull and White, 2017) suggest that the dependence of implied volatility changes on the underlying price is well captured by a low-order polynomial in $\Delta_{BS,t}$, showing $\Delta_{MV,t} - \Delta_{BS,t}$ to be quadratic function of $\Delta_{BS,t}$

Substitute into (1): $\Delta_{MV,t} = \Delta_{BS,t} + \frac{vega_{BS,t}}{S_t\sqrt{T}}(a + b\Delta_{BS,t} + c\Delta_{BS,t}^2)$

Since $dV - \Delta_{MV}dS = \varepsilon_{MV,t}$

So, $dV = \Delta_{BS,t}dS + \frac{dS}{S_t} \frac{vega_{BS,t}}{\sqrt{T}}(a + b\Delta_{BS,t} + c\Delta_{BS,t}^2) + \varepsilon_{MV,t}$

$$(dV - \Delta_{BS,t}dS) \frac{S_t}{dS} \frac{\sqrt{T}}{vega_{BS,t}} = a + b\Delta_{BS,t} + c\Delta_{BS,t}^2 + \varepsilon_{MV,t}$$

It is the regression formula.

We can rewrite it into two parts:

First, calculate the dependent variable part, $y = (dV - \Delta_{BS,t}dS) \frac{S_t}{dS} \frac{\sqrt{T}}{vega_{BS,t}} = \left(\frac{dV}{dS} - \Delta_{BS,t} \right) \frac{S_t\sqrt{T}}{vega_{BS,t}}$

then, do regression for $y = a + b\Delta_{BS,t} + c\Delta_{BS,t}^2 + \varepsilon_{MV,t}$

namely, the correction term can be calculated:

$$\Delta_{MV,t} - \Delta_{BS,t} = \frac{vega_{BS,t}}{S_t\sqrt{T}}(\hat{a} + \hat{b}\Delta_{BS,t} + \hat{c}\Delta_{BS,t}^2)$$

3. Data Preparation

Since, it is challenge to obtain the data of full option chains daily from open source, I will generate synthetic option prices data based on historical SPY prices as the underlying price, VIX index as the implied volatility base, 3-month treasury bill secondary market

rate as the risk-free rate¹, and then create moneyness and time-to-maturity two-dimensional bucketing system.

3.1 Approach to Generating Synthetic Historical Option Dataset

There are some considerations of generation of synthetic historical option price data. Initially, I created the strike grid by setting discrete strike ratio arbitrarily and use VIX index as implied volatility directly, which resulted in the binary behavior of distribution of deltas.

In order to create more balanced distributions of delta values, reflecting real-world market behaviors, volatility surface and term structure were involved. Generally, there are two methods, parametric models, such as SABR, SVI, Heston, and empirical approaches. With my preference of better theoretical control and interpretation, I incorporated the parametric volatility surface modeling to account for a skew in moneyness and term structures.

Based on my purpose, SVI model was firstly chosen in the generation of synthetic historical option dataset.

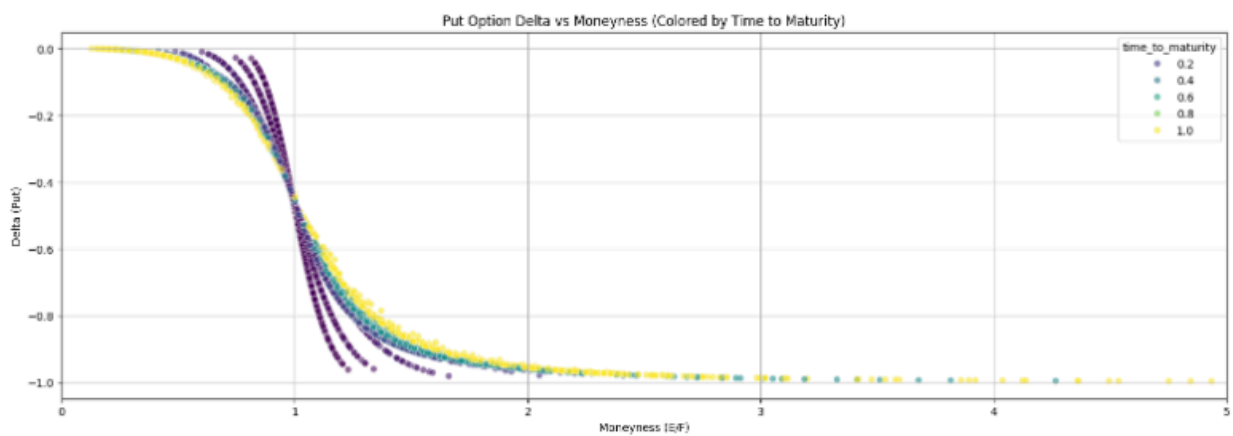
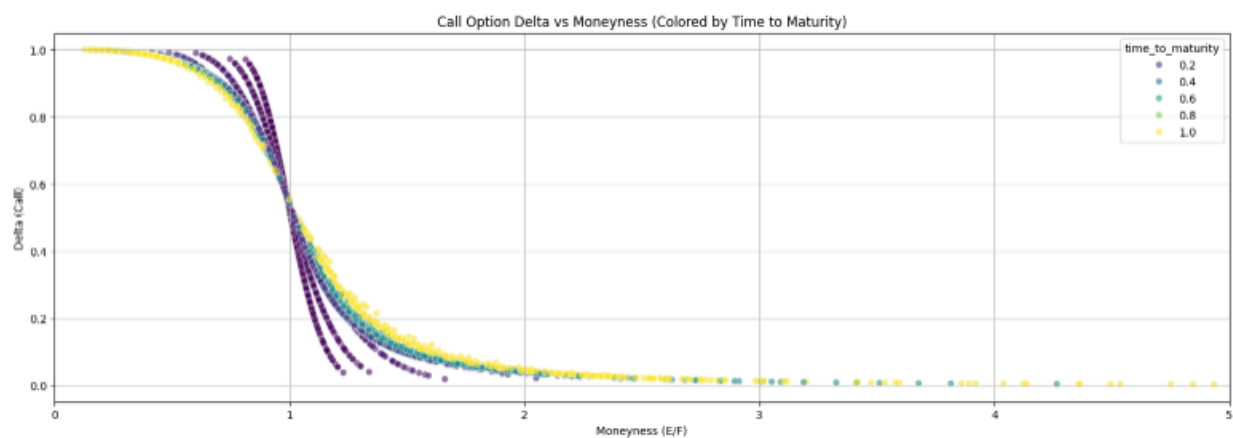
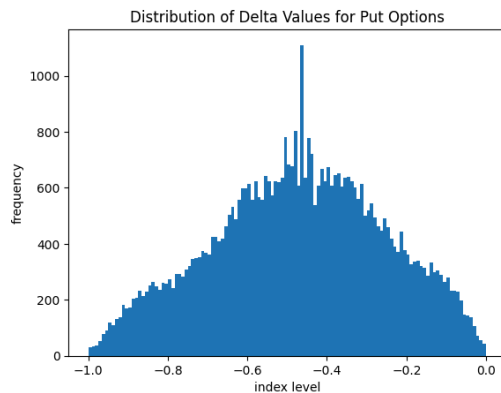
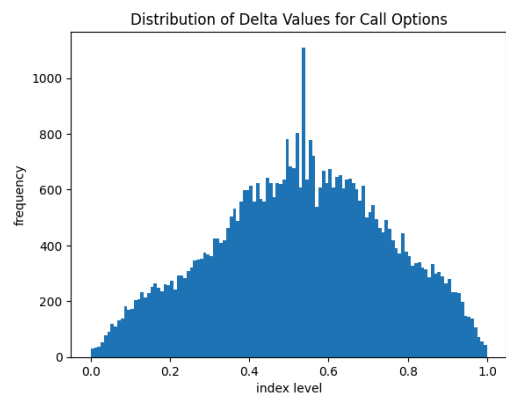
Another consideration is that the generation of the strike price grid used log-moneyness spacing with forward prices, rather than spot prices, since it is consistent with volatility surface dynamics. I have also tried to use spot prices, but it gives rise to heavily skewed distribution of delta values, particularly the long-dated options skewed toward, whereas forward-centered strike price grid can provide symmetric log-moneyness.

3.2 Statistics of the Dataset

It created 95634 option data, and 47817 are call options, and 47817 are put options.

The distribution of delta values and the relations of deltas and moneyness are showed as follow.

¹ Data source: pandas datareader, https://www.cboe.com/tradable_products/vix/ ,
<https://fred.stlouisfed.org/series/DTB3>



Call options				Put options			
Underlying price	Option price	Implied vol (SVI)	Delta	Underlying price	Option price	Implied vol (SVI)	Delta
Count	47817	47817	47817	47817	47817	47817	47817

Mean	565.170	55.07	0.565	0.529	565.170	66.309	0.565	-0.471
Std	28.072	50.78	0.197	0.217	28.072	94.827	0.197	0.217
Min	496.480	0.05	0.247	0.001	496.480	0.060	0.247	-0.999
25%	544.510	23.06	0.398	0.378	544.510	22.992	0.398	-0.622
50%	563.980	37.59	0.521	0.539	563.980	37.810	0.521	-0.461
75%	590.300	68.47	0.800	0.686	590.300	71.572	0.800	-0.314
Max	612.930	432.71	0.800	0.999	612.930	3365.337	0.800	-0.001

	Call options			Put options		
	ITM	ATM	OTM	ITM	ATM	OTM
	14517(30.36%)	18432(38.55%)	14868(31.09%)	14868(31.09%)	18432(38.55%)	14517(30.36%)
Delta distribution						
Mean	0.779	0.534	0.276	-0.221	-0.466	-0.724
Std	0.091	0.071	0.110	0.091	0.071	0.110
Min	0.628	0.354	0.001	-0.372	-0.646	-0.999
25%	0.703	0.483	0.193	-0.297	-0.517	-0.807
50%	0.766	0.540	0.293	-0.234	-0.460	-0.707
75%	0.850	0.591	0.367	-0.150	-0.409	-0.633
Max	0.999	0.695	0.477	-0.001	-0.305	-0.523

Distributions of other parameters are listed in the Appendix.

The datasets span the full deltas spectrum, with a bell-shaped, slightly skewed distribution concentrated around 0.5. The proportions of moneyness provide balanced segmentation mix, which is suitable for robust tests.

3.3 Result Discussion

3.3.1 Empirical Failure and Interpretation

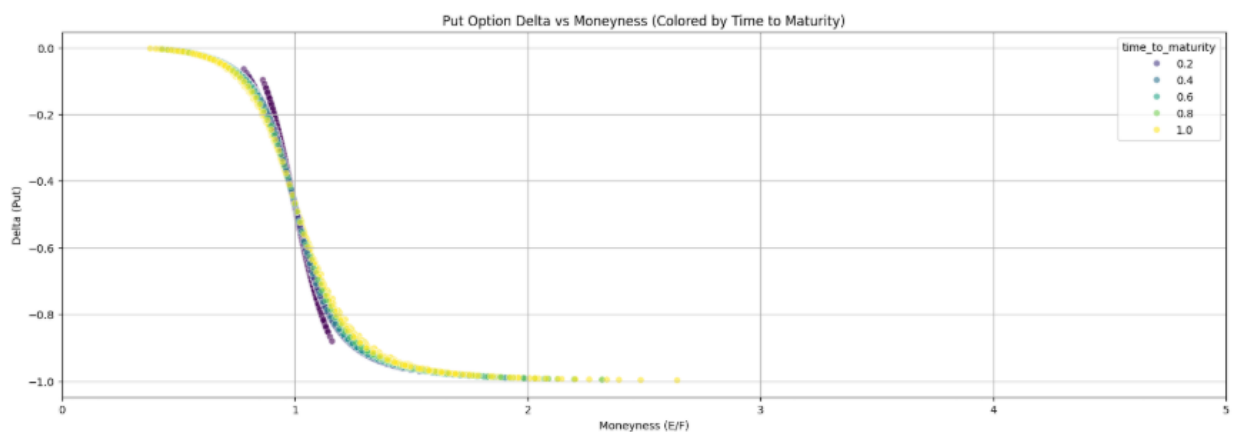
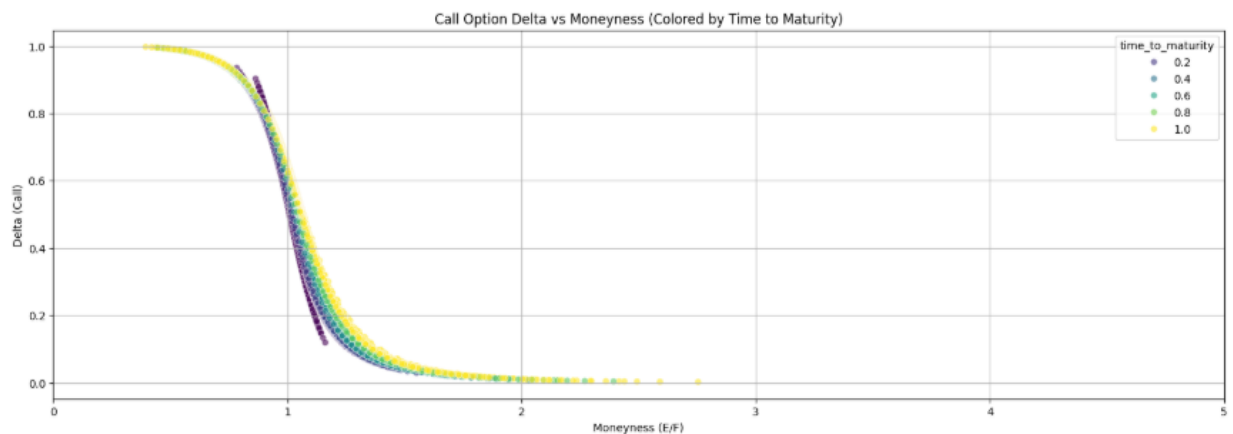
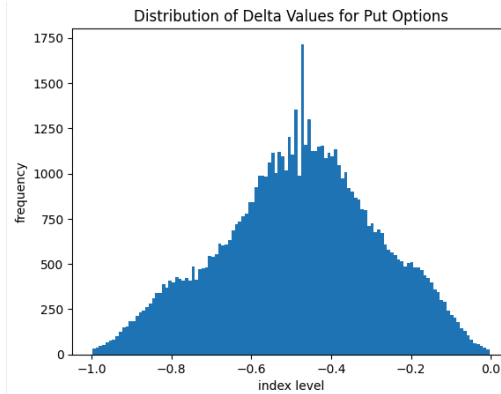
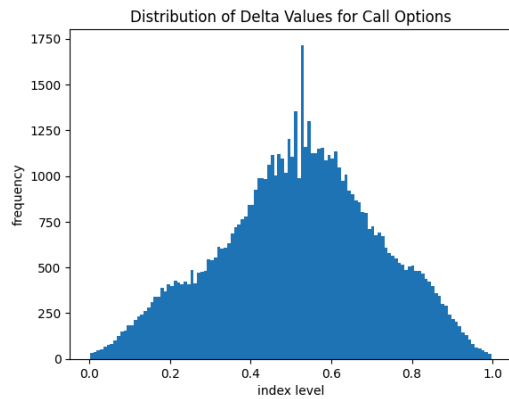
However, based on this dataset, the results have problems. Correction term and the coefficients a , b , c do not have correct magnitudes. Also, the sign of c is not consistent. Therefore, I did filtrations on dataset. Conditions are: delta $[0.15, 0.85]$, underlying price difference >0.05 , moneyness: $[0.7, 1.3]$, implied vol_svi: $[0.1, 0.6]$, vega $[1, 60]$.

However, it brings another problem that it does not leave sufficient data amount.

Therefore, I regenerate the dataset, tuning parameters in SVI equations, term structures, and spike price generation grid.

New dataset has 69069 call option data, and 69069 put option data.

The distribution of delta values and the relations of deltas and moneyness are showed as follow.



The graph of mean correction term across delta buckets shows reasonable around [0.35, 0.75] range, close to an inverted parabola-like function shape. However, the [0.75, 0.95] range may have problems.

Unfortunately, it does not solve the problem. Correction term and the coefficients a, b, c still have incorrect magnitudes.

3.3.2 Model Critique

The results above reflect that SVI model is unsuitable for data generating process for regression-based delta hedging. Its parameters are highly correlated and nonlinear curvature. Due to the nonlinear square-root structure and parameter collinearity in SVI, small perturbations in underlying asset price may induce large changes in implied volatility sensitivities, rendering $\frac{dE[\sigma_{imp}]}{dS}$ numerically unstable. However, based on the analysis above, minimum variance delta regression implicitly assumes $\frac{dE[\sigma_{imp}]}{dS}$ to be smooth, stable, and well-defined across strikes. SVI model breaks the variance minimization logic, leading to coefficients explosion and numerical instability.

Next, I will try quadratic log-moneyness formula, which ensures smoothness, scale invariance, and stable implied volatility sensitivities, which are consistent with minimum variance delta hedging theory.

3.4 Panel Dataset Generation

Here, as for the quadratic log-moneyness formula, the concrete form is as follow:

Implied_vol = base_iv + curvature_term + skew_term + noise

base_iv: control the ATM volatility regime

base_iv = base_0 * (1 + alpha * exp (- decay * T))

base_0: use VIX-index as reference

curvature_term: create smile/smirk convexity

curvature_term = (curvature_0 + curvature_01 * exp(- curvature_decay * T)) * x²

skew_term: create skew and short-dated ones have stronger skew

skew_term = skew_0 * exp(- skew_decay * T) * x

noise: add randomness

in order to avoid the dominance of noise, it needs to keep noise_std small for time-series dynamics and moneyness-dependent as well.

noise_std_eff = noise_std * (1 + wing_noise_scale * |x|)

noise = random.normal(0, noise_std_eff)

x : log-moneyness, $x = \log \frac{E}{F}$, $F = Se^{rT}$

for the analysis of minimum variance delta, the key point is the stable way of daily changes in implied volatility co-moving with daily changes in spot, namely the synthetic implied volatility dynamics, rather than the shape of the cross-sectional implied volatility smile.

Another point of generating option dataset is that in previous method the prices are generated cross-sectionally via recomputing strike prices from daily spot. It generates a daily dataset around current spot. And later, to calculate the option price differences and underlying price differences, the result is not the time difference for one option, rather than a difference between two different instruments. It may destroy the regression, since Hull-White's regression is based on one-day changes of the same option. It is a time-series hedging study. The reasonable approach is to compute differences within each option path.

Therefore, the generation should track the same option contract through time.

Concretely, each option contract is defined with issue date, strike fixed from issue date based on initial moneyness, and initial expiry, and then calculations of tracking each option contract from issue date to maturity form time-series of option prices, implied volatility and Greeks. In addition, the generations of contracts are implemented on rolling issue days. The whole process constructs the panel data structure.

4. Minimum Variance Estimation

4.1 Methodology

The previous method, that fitted quadratic formula inside delta and maturity buckets, provides another major source of instability.

The regression formula is as follow:

$$(dV - \Delta_{BS,t} dS) \frac{S_t}{dS \text{ Vega}_{BS,t}} \frac{\sqrt{T}}{S_t} = a + b\Delta_{BS,t} + c\Delta_{BS,t}^2 + \varepsilon_{MV,t}$$

regression formula: $y = a + b\Delta_{BS,t} + c\Delta_{BS,t}^2 + \varepsilon_{MV,t}$

inside each bucket, the range of $\Delta_{BS,t}$ is too small to identify a stable quadratic. Also, $\Delta_{BS,t}$ and $\Delta_{BS,t}^2$ are highly collinear. So, the regression becomes numerically fragile. Therefore, the estimation should fit on all options in the rolling window, full pooled dataset, not separately inside each delta bucket. And then, evaluations of gains are mapped to each bucket, using current time-to-maturity and observation-time delta for bucketing observations. It is the dynamic daily bucket. Bucketing is used to summarize the hedge performance and report results.

The estimate is done pooled panel regression. One subtle point is that the dataset is an unbalanced panel. Each contract has its own sequence and longer-maturity contracts contribute more, so the pooled window may be weighted implicitly toward longer-lived contracts.

Additionally, the hedge is done based on yesterday's Greeks, so the regression uses lag items of Greeks. In order to avoid numerically unstable, filtering is done on tiny dS , delta and vega.

4.2 Result Discussion

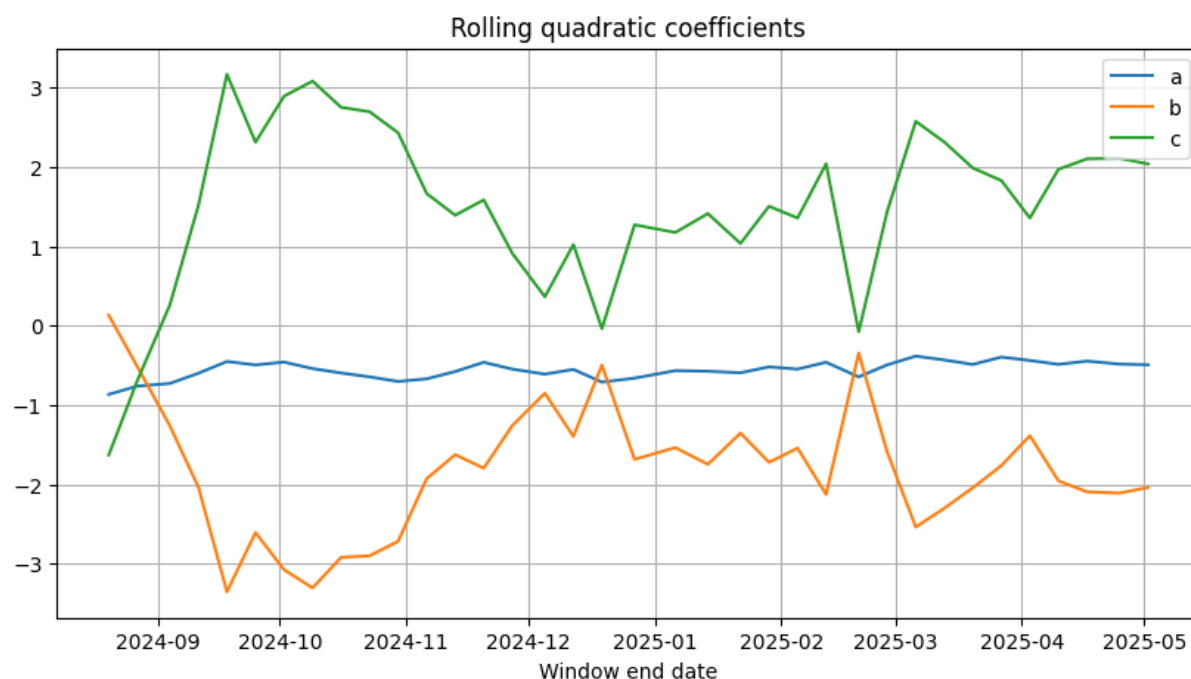
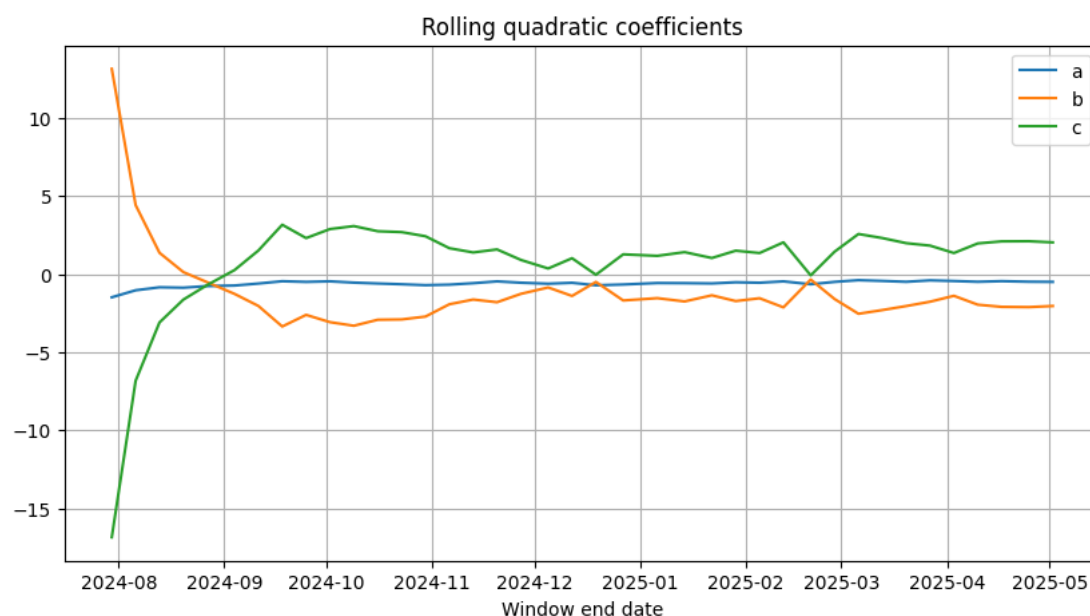
4.2.1 Coefficients

The plot of coefficients a, b, c shows that the first several rolling windows have extreme b and c, and then the coefficients quickly settle into a more stable range. The reason is that the earliest windows have too few effective observations, showing structurally immature, whereas later windows are better populated and more representative, with a richer mix of issue cohorts, remaining maturities, and delta states. By adding burn-in period, coefficients show stable behaviors.

To further contemplate behavior of b and c, the “mirror” pattern of them reflects a compensated relation between the linear and quadratic terms.

$\Delta_{BS,t}$ mostly live in a limited range, roughly between 0 and 1. On such a range, $\Delta_{BS,t}$ and $\Delta_{BS,t}^2$ are highly correlated, namely, $\Delta_{BS,t}^2$ is not very different structurally from $\Delta_{BS,t}$, so the regression can trade off b and c against each other while keeping the fitted curve nearly unchanged. Multicollinearity does not automatically invalidate the regression. The main point is to inspect the correction term: $\Delta_{MV,t} - \Delta_{BS,t} = \frac{vega_{BS,t}}{S_t\sqrt{T}} (\hat{a} + \hat{b}\Delta_{BS,t} + \hat{c}\Delta_{BS,t}^2)$, and check whether it is stable over delta buckets.

Two methods may fix this high correlation: one is to use centered delta, and then do regression on the centered items, which may break the collinearity while preserving the quadratic shape of the volatility smile. Another is to use orthogonal polynomials. Here, centering is sufficient.



Note: call options. a is intercept, b is linear term and c is quadratic coefficient

4.2.2 Gains

Another point is Gain, calculated based on the formula:

$$Gain = 1 - \frac{SSE[dV - \Delta_{MV,t}dS]}{SSE[dV - \Delta_{BS,t}dS]}$$

where SSE denotes sum of squared errors.

$SSE = \sum_t (dV_t - \Delta dS_t)^2$, Δ is $\Delta_{MV,t}$ or $\Delta_{BS,t}$

	SSE_bs	SSE_mv	Gain_pct
Raw rolling data	2081557	1011481	51.41
Burn-in period added	1742559	640638	63.23

The results show Gain_pct is around 51% and 63% with burn-in period added, which are much higher than the Gains reported by Hull-White's paper. In their paper, the overall average Gain is about 26% for calls and 23% for puts.

The results of 51% and 63% Gain_pct is a sign that the synthetic data-generating process is too favorable to minimum variance delta. The formula of synthetic implied volatility surface is a smooth deterministic function, fed with VIX-index and SPY spots, so the induced implied volatility is structured and predictable with spot movements and base volatility anchor. Also, MV delta works precisely by exploiting the predictable part of volatility change conditional on spot move. However, the dynamics of implied volatility in real markets contain substantial idiosyncratic variation and noise. The idiosyncratic variation unrelated to underlying moves may reduce MV hedging effectiveness. In addition, the relationship between the change of implied volatility and underlying movement is time-varying, namely skew and curvature change over time. Furthermore, the transaction costs and liquidity effects in real markets may widen the gap between theoretical and realized Gains.

Consequently, the higher Gain from synthetic panel data is resulted from a clean implied volatility-underlying price relationship, with a lack of time variation, idiosyncratic volatility, and microstructure noise.

4.2.3 Delta Buckets

Regarding the Gains for each delta buckets, the last bucket [0.95, 1.05) has negative Gain, reflecting the poor behavior at the extreme right end of the delta range. So, delta range will be constrained in the range [0.05, 0.95], and the extreme bucket is excluded from evaluation.

One prominent distinction between Hull-White's results and the results in this project is the pattern of hedging Gains across delta buckets. In Hull-White's result, the Gains is monotonically decreasing across delta buckets. Their hedging Gains are highest for low-delta OTM calls and lowest for high-delta OTM calls. In contrast, the results in this project show that Gains for deep ITM call options and deep OTM call options are more comparable.

A possible explanation is the maturity structure of the synthetic panel. In this project, the last two deep ITM call buckets have average maturities of approximately 0.5 and 0.47, or around 126 and 118 trading days. These options therefore retain meaningful vega and volatility sensitivity. As a result, the minimum variance delta correction remains its effectiveness of improving hedging performance in the high delta region.

In real-market datasets, very high-delta call options may often be concentrated in shorter-maturity contracts or contracts with lower remaining time values, smaller vega and more underlying-like behavior, especially when they are close to expiry ($\Delta \rightarrow 1, vega \rightarrow 0$). In that case, there is less volatility-related hedge-ratio correction for minimum-variance delta to exploit, so the hedging Gain for deep ITM calls diminishes sharply. It helps explain why the synthetic results produce more symmetric gains across low- and high- delta buckets than Hull-White's empirical results.

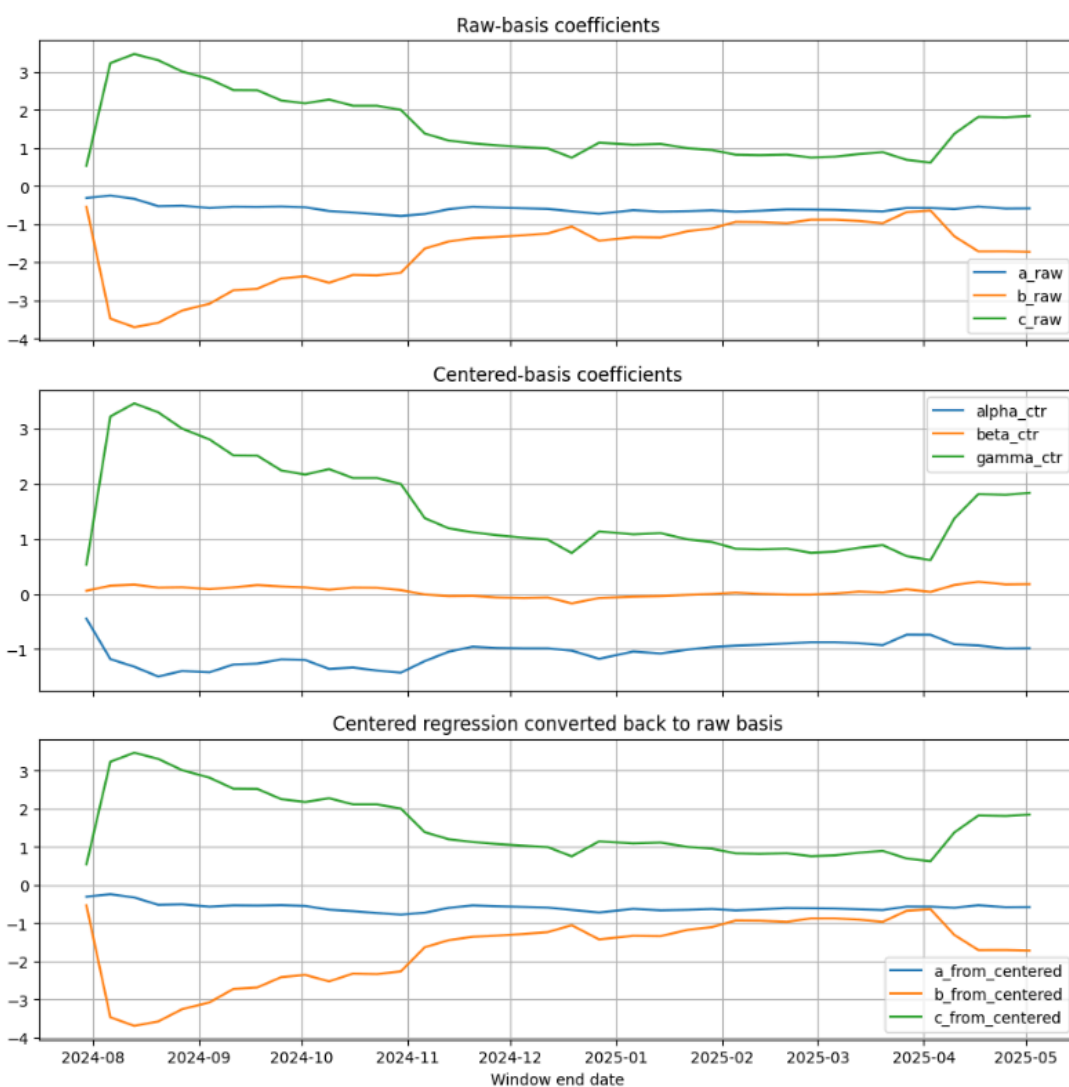
Delta_bucket	n_obs	SSE_bs	SSE_mv	Mean_delta	Mean_tau	Gain_pct
[0.05,0.15)	10,592	28,141	10,276	0.10	0.30	63.48
[0.15,0.25)	12,747	128,570	51,352	0.20	0.44	60.06
[0.25,0.35)	13,212	295,620	135,831	0.30	0.50	54.05
[0.35,0.45)	12,395	390,759	207,091	0.40	0.51	47.00
[0.45,0.55)	12,042	393,638	205,414	0.50	0.51	47.82
[0.55,0.65)	12,644	363,150	168,357	0.60	0.50	53.64
[0.65,0.75)	13,735	279,621	121,349	0.70	0.50	56.60

[0.75,0.85)	16,791	156,463	71,351	0.80	0.50	54.40
[0.85,0.95)	25,328	41,542	33,701	0.90	0.47	18.87
[0.95,1.05)	28,705	2,739	6,271	0.98	0.28	-128.87

4.2.4 Regression on Raw Basis and Centered-basis

The following plots show that centered-basis delta does not change the fitted curve.

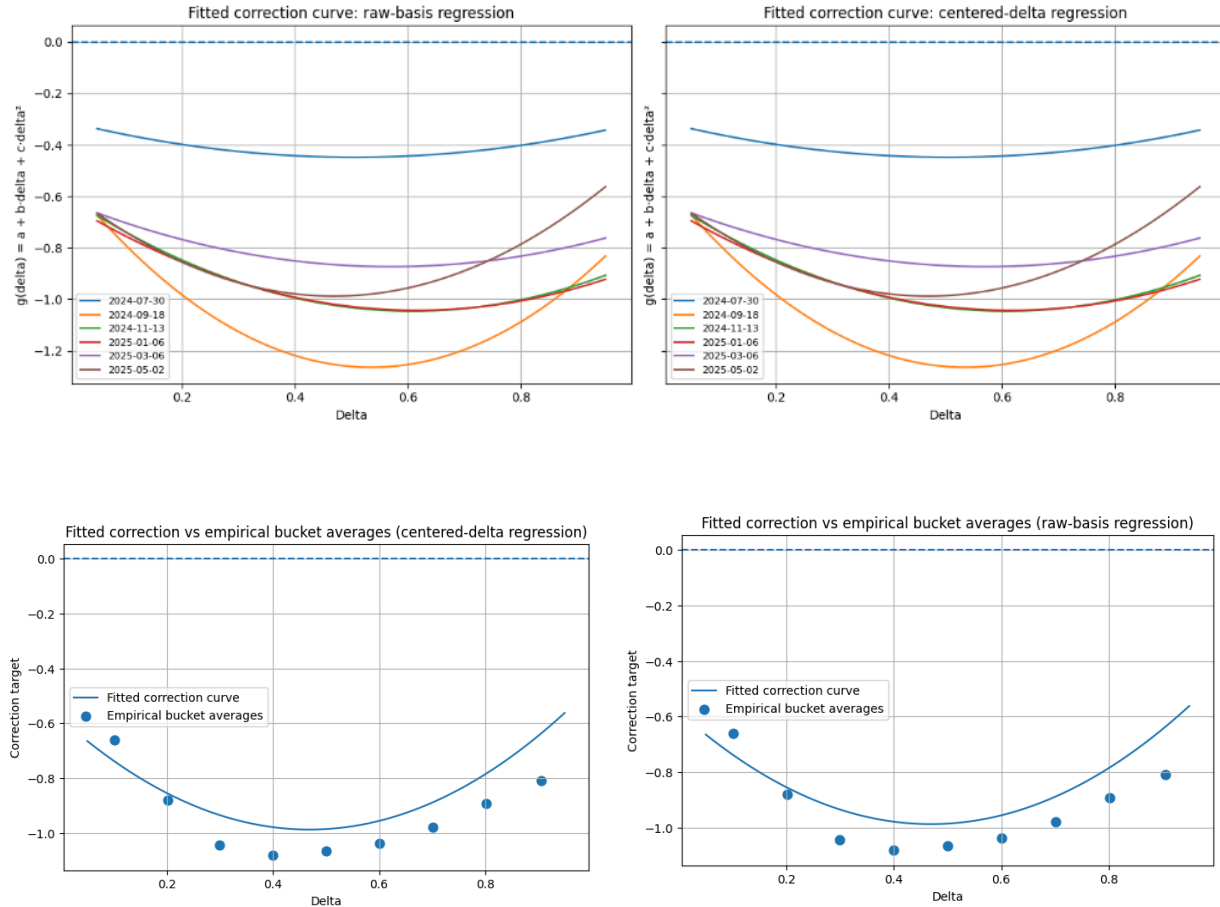
Centered-delta regression is numerically more stable. Intercept is stable around -1, and roughly within the range [-1.5, -0.5]. Slope is near zero, slightly positive, which suggests that linear term is largely an artifact of correlation with the quadratic term in raw basis.



Notes: call options. a is intercept, b is linear term, c is quadratic coefficient. Alpha is intercept, beta is linear term, gamma is quadratic coefficient.

4.2.5 Fitted Correction Curve and Empirical Bucket-level Average Correction

The fitted correction curves display that the fitted correction terms are negative across all the delta range, roughly U-shaped, lowest around mid-deltas and less negative toward low and high deltas. Also, with a trimmed range of delta, the distortion of right tail is eliminated. Therefore, the fitted correction function is stable and interpretable, and the main central delta region is captured reasonably well.



5. Future Work

The results in this report are based on the rigid, deterministic formula of the volatility surface generator. With the purpose of testing real-world risk simulations, this determinism is not ideal, since it simplifies the real market and may be considered as reverse-engineering. Hence, a meaningful method is to embed the structural traits to evolve over time, such as autoregression terms.

Appendix

Original dataset

Call options

	underlying_price	strike_price	time_to_maturity	days_to_expire	implied_vol_svi	risk_free_rate	option_price	delta	vega	d1
count	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000
mean	565.170474	11145.714676	0.294218	74.142857	0.565436	4.630158	55.070866	0.528522	80.420890	0.084291
std	28.072129	26461.478886	0.325852	82.114767	0.196795	0.429537	50.780138	0.216860	46.072254	0.661568
min	496.480000	423.746688	0.019841	5.000000	0.247131	4.170000	0.053183	0.000768	0.938507	-3.167943
25%	544.510000	703.841959	0.039683	10.000000	0.397717	4.220000	23.064583	0.377841	43.956496	-0.311157
50%	563.980000	1223.563825	0.166667	42.000000	0.521310	4.480000	37.590671	0.539163	69.392100	0.098325
75%	590.300000	4582.586198	0.500000	126.000000	0.800000	5.140000	68.467611	0.686082	101.989671	0.484776
max	612.930000	381063.617102	1.000000	252.000000	0.800000	5.260000	432.710861	0.999466	244.494329	3.271718

Put options

	underlying_price	strike_price	time_to_maturity	days_to_expire	implied_vol_svi	risk_free_rate	option_price	delta	vega	d1
count	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000	47817.000000
mean	565.170474	11145.714676	0.294218	74.142857	0.565436	4.630158	66.308619	-0.471478	80.420890	0.084291
std	28.072129	26461.478886	0.325852	82.114767	0.196795	0.429537	94.827005	0.216860	46.072254	0.661568
min	496.480000	423.746688	0.019841	5.000000	0.247131	4.170000	0.059833	-0.999232	0.938507	-3.167943
25%	544.510000	703.841959	0.039683	10.000000	0.397717	4.220000	22.991970	-0.622159	43.956496	-0.311157
50%	563.980000	1223.563825	0.166667	42.000000	0.521310	4.480000	37.810137	-0.460837	69.392100	0.098325
75%	590.300000	4582.586198	0.500000	126.000000	0.800000	5.140000	71.571856	-0.313918	101.989671	0.484776
max	612.930000	381063.617102	1.000000	252.000000	0.800000	5.260000	3365.336595	-0.000534	244.494329	3.271718

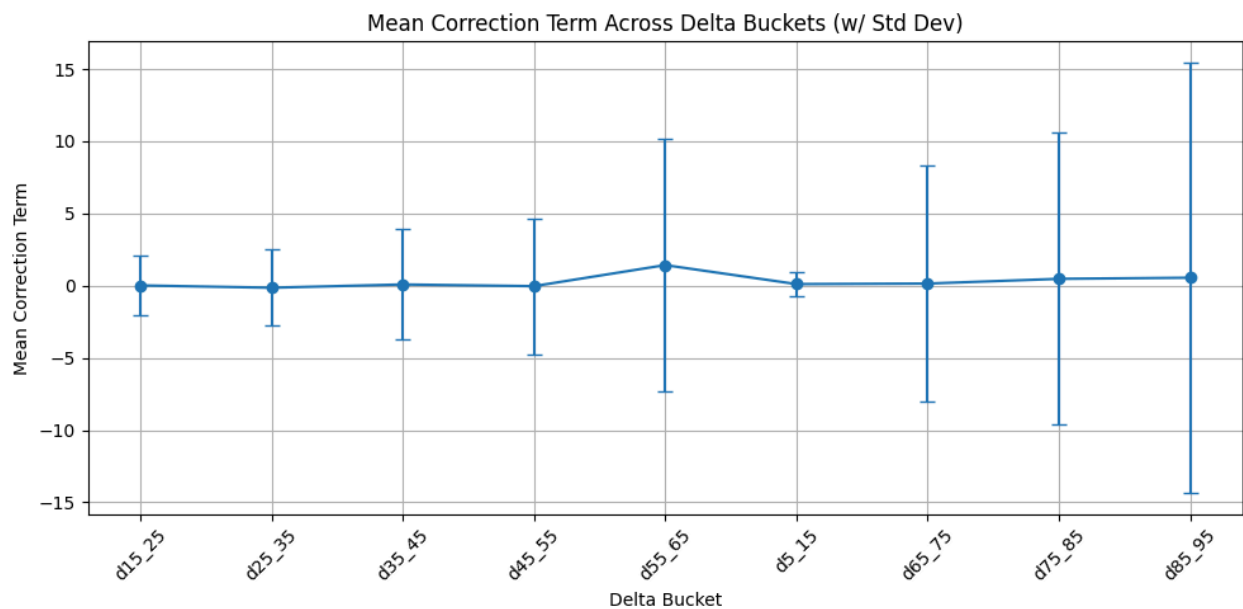
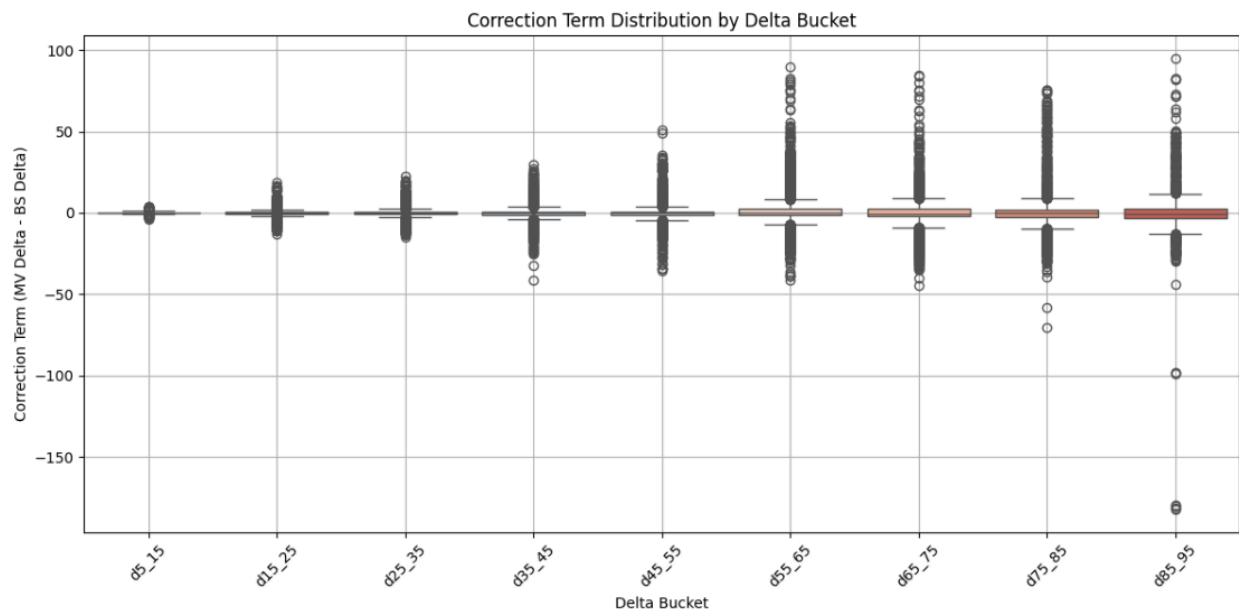
New dataset

Call options

	underlying_price	strike_price	time_to_maturity	days_to_expire	implied_vol_svi	risk_free_rate	option_price	delta	vega	d1	Forward_price	moneyness
count	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000
mean	565.170474	578.901312	0.398526	100.428571	0.325396	0.046302	37.648892	0.520498	106.247258	0.056199	565.274464	1.024278
std	28.072039	68.062184	0.337966	85.167440	0.148750	0.004295	28.853382	0.196441	51.598145	0.578624	28.071032	0.113015
min	496.480000	195.975388	0.039683	10.000000	0.156513	0.041700	0.082516	0.001946	3.070086	-2.886832	496.488255	0.394564
25%	544.510000	538.073350	0.083333	21.000000	0.193936	0.042200	19.511543	0.391663	63.548522	-0.274988	544.581199	0.967061
50%	563.980000	573.601624	0.250000	63.000000	0.276409	0.044800	30.021899	0.526744	95.560949	0.067087	564.039080	1.011390
75%	590.300000	612.354792	0.750000	189.000000	0.463400	0.051400	46.284981	0.656065	143.437968	0.401748	590.495274	1.064879
max	612.930000	1367.715790	1.000000	252.000000	0.600000	0.052600	308.721865	0.997198	244.506663	2.770091	613.188711	2.753672

Put options

	underlying_price	strike_price	time_to_maturity	days_to_expire	implied_vol_svi	risk_free_rate	option_price	delta	vega	d1
count	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000	69069.000000
mean	565.170474	578.901312	0.398526	100.428571	0.325396	0.046302	40.715911	-0.479502	106.247258	0.056199
std	28.072039	68.062184	0.337966	85.167440	0.148750	0.004295	38.345551	0.196441	51.598145	0.578624
min	496.480000	195.975388	0.039683	10.000000	0.156513	0.041700	0.175535	-0.998054	3.070086	-2.886832
25%	544.510000	538.073350	0.083333	21.000000	0.193936	0.042200	19.654493	-0.608337	63.548522	-0.274988
50%	563.980000	573.601624	0.250000	63.000000	0.276409	0.044800	30.101322	-0.473256	95.560949	0.067087
75%	590.300000	612.354792	0.750000	189.000000	0.463400	0.051400	47.189100	-0.343935	143.437968	0.401748
max	612.930000	1367.715790	1.000000	252.000000	0.600000	0.052600	815.195008	-0.002802	244.506663	2.770091



	a	b	c
count	2461.000000	2461.000000	2461.000000
mean	-425.532891	882.749306	-197.242855
std	26977.267043	64103.171493	39483.003421
min	-357658.378703	-869762.525918	-463101.007519
25%	-1104.324022	-3201.156398	-4616.556170
50%	3.767346	-68.191594	129.794246
75%	666.486051	4384.999766	4047.717568
max	376705.101911	814189.913865	501972.187560